Math 141, S10 Name: Answer

Quiz 3 Section: SSN: xxx-xx-

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(6 points) Evaluate each of the following limits if it exists.

(a) $\lim_{x\to 3} \frac{1}{\sqrt{x^3-6x}}$ $= \frac{\lim_{x\to 3} 1}{\lim_{x\to 3} \sqrt{x^3-6x}} = \frac{1}{\sqrt{3^3-6\cdot 3}}$

$= \frac{1}{\sqrt{9}} = \frac{1}{3}$

(b) $\lim_{x\to 2} \frac{x^2+x-6}{x-2}$ $= \lim_{x\to 2} \frac{\cancel{(x-2)}(x+3)}{\cancel{x-2}} = \lim_{x\to 2} (x+3) = 2+3 = 5$

(c) $\lim_{x\to+\infty} \frac{3x^2+1}{2x^2-1}$ $= \lim_{x\to+\infty} \frac{\frac{3x^2+1}{x^2}}{\frac{2x^2-1}{x^2}} = \lim_{x\to+\infty} \frac{3+\frac{1}{x^2}}{2-\frac{1}{x^2}}.$

$= \frac{\lim_{x\to+\infty} (3+\frac{1}{x^2})}{\lim_{x\to+\infty} (2-\frac{1}{x^2})} = \frac{3}{2}$

Problem 1.(4 points) Explain why this function is discontinuous at $x = 0$:

$$f(x) = \begin{cases} \cos(x) & x \geq 0, \\ 0 & x < 0. \end{cases}$$

$\lim_{x\to 0^-} f(x) = 0$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \cos x = \cos 0 = 1$

$\Rightarrow \lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$,

thus $f(x)$ is discontinuous at $x=0$.